

Predicting Solar Output to Prescribe Battery Dispatch Decisions

Robust Optimization for the San Diego Electricity Market

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1 Introduction

The power market in Southern California, specifically in San Diego, experiences pronounced price volatility due to high midday solar irradiance and sharp evening demand peaks, frequently resulting in negative Locational Marginal Prices (LMPs) and renewable curtailment. As a result, these climate and grid conditions make it an ideal setting for evaluating the operational value of solar-paired battery assets. In this environment, the timing of battery charging, discharging, or holding decisions plays a critical role in mitigating curtailment and capturing value from price fluctuations.

However, the operational performance of these assets depends heavily on accurate short-term forecasts and intelligent dispatch strategies. Solar generation is inherently uncertain due to weather variability, while day-ahead electricity prices in the San Diego Gas & Electric (SDG&E) region exhibit sharp fluctuations driven by solar oversupply during daylight hours and peak demand in the evening. Myopic heuristic dispatch rules or inaccurate forecasts can lead to missed charging opportunities, inefficient discharging, and lost economic value.

To address this challenge, we develop a predictive-to-prescriptive framework that combines optimal solar generation forecasting with an optimization-based battery dispatch model. Using day-ahead electricity prices in the SDG&E market, the model prescribes optimal battery actions both nominally and in regards to accounting for uncertainty in price forecasts through robust optimization formulations.

2 Data

National Solar Radiation Database (NSRDB)

For the first aspect of this framework, predicting next day solar GHI, we needed a dataset that captured local weather patterns, climate information, and solar irradiance at an hourly level. The National Solar Radiation Database is a public database that contained gridded estimates of hourly solar irradiance, wind speed, sun zenith, cloud cover, humidity, and other climate indicators [1].

We downloaded this dataset for the entire San Diego region that is covered by San Diego Gas & Electric. Since this database contains a unique dataframe per grid cell location, we averaged each cell (each weather-related variable) of all the location-specific dataframes to get one general dataset for the entire San Diego area. This dataset contained 1 year worth of climate information, with each observation representing 1 hour. We engineered our target variable to be the next day solar irradiance, so we could predict next-day solar levels to plan our battery schedule.

CAISO Day-Ahead Locational Marginal Pricing

CAISO (California Independent System Operator) manages the bulk power grid and wholesale electricity markets for most of California. They publish hourly day-ahead locational marginal prices (LMPs), which is the cost of supplying one additional MWh of electricity at a specific location during a certain hour. These prices are the key signal for storage and solar operators to charge or discharge batteries [2].

We filtered this data to only the SDG&E nodes in the 2024 CAISO day-ahead LMP dataset. This ensured all analysis reflects market conditions specific to the covered San Diego region.

3 Methodology Overview

Our prediction-to-prescription framework integrated next-day solar generation forecasts with electricity price data to determine optimal battery dispatch decisions. We used hourly observations that spanned all of 2024 and adopt a time-based split, training all prediction models on the first 51 weeks of data and evaluating performance on the final week. This design mirrored the operational setting in which next-day decisions rely solely on historically available information.

Next-Day Solar Irradiance Prediction

To generate solar energy forecasts for battery scheduling, we trained a solar irradiance prediction model using historical data from the National Solar Radiation Database. The predictor variables included local weather variables, and cyclic time features to capture daily and seasonal patterns. With the work of some feature-engineering, the models output hourly next-day Global Horizontal Irradiance (GHI next day) predictions. Among the models evaluated, an Optimal Regression Tree achieved the lowest mean squared error and was therefore selected to provide solar inputs to the battery-dispatch optimization.

Battery Dispatch Optimization

We modeled a solar-paired battery operating within the SDG&E network, where charging is restricted to on-site solar generation and direct grid purchases are not permitted. This reflects a common CAISO configuration for solar-paired storage that does not operate as a price-arbitrage merchant. Using the predicted hourly GHI from the Optimal Regression Tree, we formulated an optimization model that determines the battery’s charging, discharging, and grid-export decisions subject to operational constraints. The model is solved on a rolling daily basis, with each day’s dispatch optimized using day-ahead price information and the battery’s end-of-day state of charge carried forward to the following day. This formulation translates forecasted solar supply into a feasible and economically efficient next-day dispatch schedule consistent with SDG&E operating conditions.

Robust Optimization for Price Uncertainty

Day-ahead electricity prices are uncertain at the time dispatch decisions are made. To account for this uncertainty, we extend the dispatch model to include robust optimization formulations that protect against adverse deviations from forecasted SDG&E node Location Marginal Prices. These formulations produce dispatch schedules that remain economically stable under price forecast errors while preserving the structure of the deterministic model, with robustness applied within each daily re-optimization using day-ahead price information.

4 Solar Irradiance Forecasting

Feature Engineering

To support next-day solar irradiance prediction, we constructed a feature set combining historical irradiance patterns, local meteorological conditions, and time-based signals known to influence solar generation. As mentioned previously, the underlying dataset was sourced from the National Solar Radiation Database (NSRDB), which provides hourly gridded estimates of key climate variables including Global Horizontal Irradiance (GHI), Direct Normal Irradiance (DNI), Diffuse Horizontal Irradiance (DHI), temperature, humidity, wind speed, and atmospheric pressure. Because the SDG&E region spans multiple NSRDB grid cells, we downloaded all corresponding files and averaged each variable across cells to produce a single hourly dataset for the region. These features enable the model to learn both physical drivers of irradiance and cyclical nature of solar production. The dataset was sorted chronologically and split using a time-based partition, ensuring that the model trained only on past observations and was evaluated on the latest periods.

To align the data with our forecasting objective, we also engineered the target variable as the next-day irradiance value $GHI_{next\ day}_t = GHI_{t+24}$ which was implemented via a 24-hour forward shift. This links each hour’s weather conditions to the solar output observed exactly one day later.

Model Selection

We evaluated several forecasting models, including linear regression, k-nearest neighbors, decision trees, Random Forests, and XGBoost, to identify a suitable predictor for next-day Global Horizontal Irradiance

(GHI).

Model	Key settings	Test MSE
Linear Regression	–	1724.22
k -NN Regressor	$k = 39$ (tuned)	1978.47
Decision Tree	max depth = 4, min leaf = 50	2244.79
Random Forest	300 trees, min leaf = 5	1912.59
Gradient Boosting (XGBoost-style)	500 trees, $\eta = 0.03$, depth = 3	1729.96
Optimal Regression Tree (ORT)	depth = 5	1711.68

Table 1: Test mean squared error (MSE) for candidate next-day GHI forecasting models, including the final Optimal Regression Tree (ORT) with depth 5.

From this table we see, linear regression performed competitively, suggesting that much of the variation in next-day irradiance was explained by approximately linear relationships with meteorological and time-based features. Nonlinear models such as kNN and standard decision trees were more sensitive to noise and performed comparatively worse. Random Forests and XGBoost improved stability but did not substantially outperform simpler baselines in terms of test MSE.

The Optimal Regression Tree (ORT) with depth 5 achieved the lowest test MSE across all models. This ORT relied on a small number of physically meaningful variables to partition the data into interpretable irradiance measures. The highest-level splits were driven by certain irradiance measures (ex. GHI and clear-sky GHI), which separated low-irradiance periods from high solar potential conditions. Subsequent splits refined these predictions using time-based features such as hour and month which captured daily and seasonal effects that strongly influence solar availability. Within each branch, the tree produced stable prediction ranges corresponding to distinct weather–time combinations, rather than overfitting to local noise. Therefore, we decided to use this ORT as our final model for solar predictions.

5 Deterministic Solar-Battery Optimization Model

We then used a deterministic day-ahead optimization model for a solar-coupled battery operating over a discrete hourly horizon $t = 1, \dots, T$. The objective was to schedule hourly charging, discharging, and exports for each operating day in a way for the horizon that maximizes energy-market revenue while respecting all physical and operational constraints of the system. The model is re-solved daily using day-ahead price information, with the battery’s end-of-day state of charge carried forward to the next day. As mentioned, the battery cannot purchase electricity from the grid, so solar availability and storage decisions fully determine its ability to supply energy.

Parameters

The model relied on both parameter forecasts and fixed battery specifications. The day-ahead market price p_t represents the revenue that can be earned for each MWh exported at time t , while the predicted solar energy $\hat{s}_t$ indicates how much renewable energy is available to either charge the battery or be exported directly. The battery itself is characterized by its energy capacity $E_{\max}$ and its maximum charging and discharging power limits $P_{\max}^c$ and $P_{\max}^d$, which restrict the rate at which energy can enter or leave storage. Charging and discharging efficiencies (η_c, η_d) model round-trip conversion losses, and the initial state of charge SoC_0 specifies the energy level at the start of the period. A degradation cost k_{deg} is applied to discharged energy, reflecting wear on the battery, and each time step has length $\Delta t = 1$ hour.

p_t	Day-ahead locational marginal price at time t [\$/MWh]
$\hat{s}_t$	Solar energy prediction at time t [MWh]
$E_{\max} = 20$ MWh	Battery energy capacity [MWh]
$P_{\max}^c = 5.0$ MW	Maximum charging power [MW]
$P_{\max}^d = 5.0$ MW	Maximum discharging power [MW]
$\eta_c = 0.95$	Charging efficiency
$\eta_d = 0.95$	Discharging efficiency
$\text{SoC}_0 = 0$ MWh	Initial state of charge [MWh]
$k_{\text{deg}} = \$5.0/\text{MWh}$	Degradation cost per MWh discharged
$\Delta t = 1.0$ hr	Length of each time step [hours]

Decision Variables

At each hour, the model chooses operational levels describing how the battery interacts with available solar energy and the grid. The charging variable c_t represents the portion of solar production stored in the battery, while the discharging variable d_t quantifies energy discharged from storage to be sent to the grid. The state of charge SoC_t tracks stored energy over time and must remain within the feasible limits of the battery. Finally, the export variable e_t captures the energy delivered to the grid which consists of both direct solar and any discharged energy.

$$\begin{aligned}
c_t &\geq 0 && \text{Battery charging power [MW]} \\
d_t &\geq 0 && \text{Battery discharging power [MW]} \\
\text{SoC}_t &&& \text{State of charge at the end of hour } t \text{ [MWh]} \\
e_t &\geq 0 && \text{Energy exported to the grid [MW]}
\end{aligned}$$

Objective

The objective maximizes revenue from exported energy while accounting for degradation costs. Since the battery sells energy at price p_t when exporting e_t , and incurs a degradation cost proportional to discharged energy, the total profit over the horizon is:

$$\max_{\{c_t, d_t, \text{SoC}_t, e_t\}} \sum_{t=1}^T \Delta t (p_t e_t - k_{\text{deg}} d_t). \quad (1)$$

This formulation encourages the battery to store solar energy during lower-price periods and discharge when prices increase but through an optimization lens rather than a heuristic approach, capturing the fundamental economics of energy shifting.

Constraints

Our objective is subject to the follow constraints:

$$\begin{array}{ll}
\text{SoC}_0 = 0 & t = 0 \quad (\text{Initial condition}) \\
\text{SoC}_t = \text{SoC}_{t-1} + \eta_c c_t \Delta t - \frac{1}{\eta_d} d_t \Delta t & t = 1, \dots, T \quad (\text{Battery dynamics}) \\
0 \leq \text{SoC}_t \leq E_{\max} & t = 0, \dots, T \quad (\text{Energy capacity}) \\
0 \leq c_t \leq P_{\max}^c & t = 1, \dots, T \quad (\text{Charging limits}) \\
0 \leq d_t \leq P_{\max}^d & t = 1, \dots, T \quad (\text{Discharging limits}) \\
c_t \leq \hat{s}_t & t = 1, \dots, T \quad (\text{No grid charging}) \\
e_t = \hat{s}_t - c_t + d_t & t = 1, \dots, T \quad (\text{Export balance}) \\
e_t \geq 0 & t = 1, \dots, T \quad (\text{Nonnegativity})
\end{array}$$

Our battery feasibility is enforced through this set of physical and operational constraints. The initial state of charge at the beginning of the horizon is fixed at 0, and storage evolves through a linear dynamic equation that incorporates efficiency-adjusted charging and discharging. The state of charge must lie within the range $[0, E_{\max}]$ at all times. Charging and discharging power are each limited by their respective maximum capabilities, and because the battery cannot charge from the grid, charging is constrained by available solar production. Finally, an export balance equation ensures that all exported energy originates from solar generation or battery discharge.

6 Robust Optimization Under Price Uncertainty

As mentioned previously, day-ahead market prices in the SDG&E system are inherently uncertain, often shifting in response to weather-driven renewable output, load fluctuations, and congestion patterns. Since our battery must commit to a fixed next-day dispatch plan each operating day and cannot buy from the grid, it is particularly exposed to errors in these price forecasts as overestimating prices may lead the battery to discharge too early, while underestimating prices may cause missed revenue opportunities. To mitigate this vulnerability, we explored robust optimization approaches that explicitly account for potential price deviations. Incorporating uncertainty into the model allows the dispatch to prioritize strategies that remain profitable even when realized prices differ from expectations. This produces schedules that are more stable, less sensitive to forecast error, and better aligned with real-world operating conditions.

Ellipsoidal Price Uncertainty

To address the sensitivity of next-day battery schedules to inaccurate price forecasts, we incorporate ellipsoidal uncertainty into the revenue term of the optimization model. Ellipsoidal uncertainty provides a structured way to guard against these errors by allowing prices to vary within a continuous uncertainty set centered around the forecast, capturing correlated forecast errors across the daily dispatch horizon.

To model the ellipsoidal price uncertainty, we incorporate it into the revenue component of the optimization model. We model the true price vector as

$$p = \hat{p} + u, \quad \|u\|_2 \leq \Gamma_p,$$

where $\hat{p}$ denotes the day-ahead forecast and u represents deviations constrained Γ_p . Solving the inner minimization problem yields the robust objective:

$$\max \left\{ \sum_{t=1}^T \Delta t (\hat{p}_t e_t - k_{\text{deg}} d_t) - \Gamma_p \|e\|_2 \right\}$$

The additional term $\Gamma_p \|e\|_2$ introduces a second-order cone penalty on battery schedules that rely heavily on exporting during hours with high price uncertainty, thereby discouraging extreme, price-dependent behavior. This results in more stable next-day dispatch strategies that remain profitable even when

actual prices deviate from forecasts. In our experiments, we set $\Gamma_p = 10$, corresponding to a moderate level of aggregate price uncertainty over the daily dispatch horizon. Importantly, all operational battery constraints remain unchanged, only the objective function is modified to reflect robustness with respect to price uncertainty.

Budgeted Price Uncertainty

As an alternative to the ellipsoidal formulation, we also consider a budgeted uncertainty set, which provides a more conservative form of worst-case price protection. In this framework, hourly prices are allowed to experience adverse (downward) deviations from their nominal forecasts, but only a limited number of such deviations may simultaneously attain their worst-case values. This is modeled as

$$p_t = \hat{p}_t - u_t, \quad 0 \leq u_t \leq \delta_t, \quad \sum_{t=1}^T \frac{u_t}{\delta_t} \leq \Gamma_p,$$

where δ_t specifies the maximum downward deviation in hour t and Γ_p controls the total magnitude of adverse price realizations the model guards against.

In our implementation, we set $\delta_t = 0.20 \hat{p}_t$, such that prices may deviate downward by up to 20% of the nominal forecast in any hour. We set the budget parameter to $\Gamma_p = 5$, which corresponds to protecting against a worst-case scenario where approximately five hours within the daily horizon experience their maximum adverse deviation simultaneously (with the remaining hours experiencing smaller or no deviations).

Under this uncertainty structure, the robust counterpart of the revenue term becomes

$$\min_u \sum_{t=1}^T \Delta t (\hat{p}_t - u_t) e_t \quad \text{s.t. the budgeted constraints above,}$$

which yields a linear robust objective after taking the dual:

$$\max \left\{ \sum_{t=1}^T \Delta t (\hat{p}_t e_t - k_{\text{deg}} d_t) - \Gamma_p \theta - \sum_{t=1}^T \Delta t q_t \right\},$$

subject to:

$$q_t \geq 0, \quad q_t \geq e_t \delta_t - \theta, \quad t = 1, \dots, T,$$

where q_t and θ are auxiliary dual variables.

Compared to ellipsoidal uncertainty, the budgeted model offers stronger, more conservative protection by explicitly guarding against the worst possible subset of downward price shocks up to size Γ_p . This yields dispatch schedules that are more conservative but more resilient to extreme deviations from prices. As before, all physical battery constraints remain unchanged.

7 Baseline Rule-Based Dispatch Strategy

Heuristic Design

In practice, many battery operators rely on simple rule-based heuristics to determine charging and discharging behavior. While these approaches can capture broad market signals and are easy to implement, they do not produce optimal dispatch schedules. Rule-based heuristics typically make decisions myopically at each time step rather than optimizing jointly over the full planning horizon. As a result, they fail to account for time-based trade-offs in state of charge and price dynamics, leading to suboptimal scheduling decisions and consequently, lower realized revenue compared to optimization-based strategies.

Behavioral Characteristics

To benchmark our optimization models, we consider two heuristic baselines that reflect common operational behavior. The first baseline immediately sells all available solar generation to the grid, effectively following a price-weighted irradiance strategy with no battery/storage capabilities. While this approach avoids operational complexity, it ignores time-based price variation and forfeits the opportunity to shift energy from low-price to high-price periods.

The second baseline follows a simple threshold-based charging and discharging rule: the battery charges during daylight hours when prices fall in the bottom 10% of the daily price distribution and discharges when prices exceed the top 90%. This heuristic reflects an attempt to exploit daily price cycling but relies on fixed quantile thresholds and does not adapt to changing market conditions or multi-day dynamics. Consequently, it often results in rigid daily cycling patterns, mistimed charging or discharging events, and inefficient use of battery capacity. These limitations highlight the inability of heuristic rules to fully capture the structure of intraday price variability and intertemporal storage value.

8 Results

Weekly Profit Comparison Across Strategies

Scenario	Profit (\$)
No Battery (Baseline 1)	2595.45
Rule-Based Dispatch (Baseline 2)	3207.59
Nominal Optimization	3849.04
Ellipsoidal Robust Optimization	3465.84
Budget Robust Optimization	3285.86

Table 2: Weekly profit comparison across dispatch strategies.

Across the week, adding storage improves profitability relative to directly selling solar to the grid (Table 2). The no-battery baseline earns \$2,595.45, while the rule-based dispatch heuristic increases profit to \$3,207.59 by shifting some generation away from low-price hours. The highest weekly profit is achieved by nominal optimization (\$3,849.04), illustrating the value of optimizing charging and discharging jointly over the full horizon rather than relying on myopic threshold rules.

Introducing price robustness reduces profit relative to the nominal solution, reflecting the cost of hedging against price uncertainty. Ellipsoidal robust optimization achieves \$3,465.84, while the more conservative budgeted robust model yields \$3,285.86. Importantly, both robust strategies still outperform the heuristic and no-storage baselines, indicating that robustness preserves a meaningful profit uplift while providing downside protection.

Dispatch Behavior Comparison

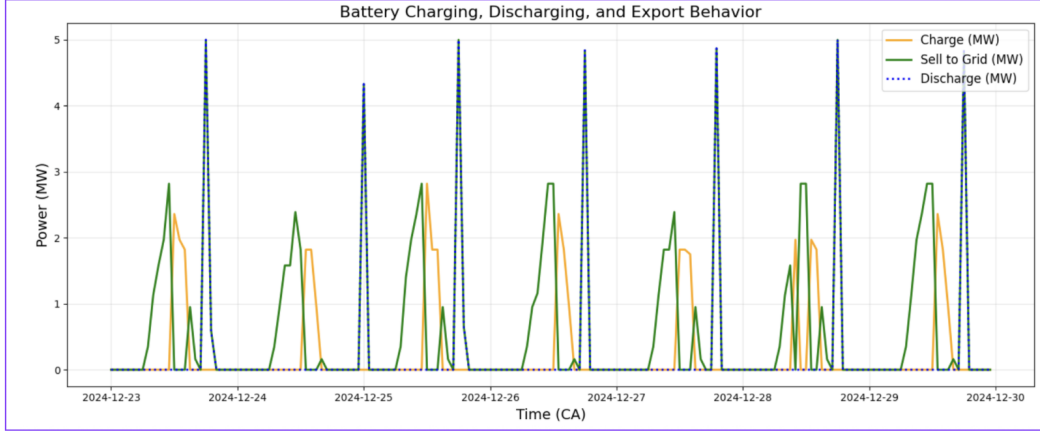


Figure 1: Rule-based baseline dispatch behavior over the 7-day horizon.

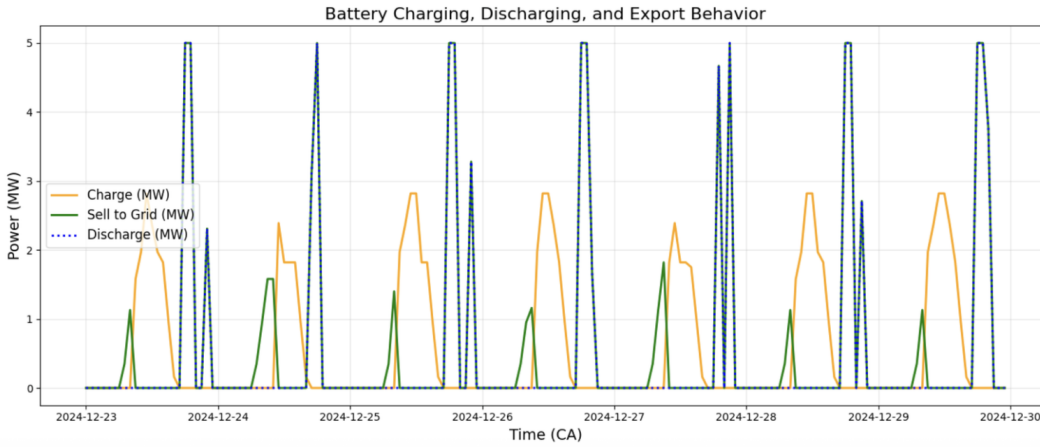


Figure 2: Nominal optimization dispatch behavior over the 7-day horizon.

To understand the drivers of these profit differences, Figures [1](#) and [2](#) compare dispatch behavior over the same 7-day horizon under the rule-based heuristic and the nominal optimizer. The heuristic produces rigid daily patterns governed by fixed thresholds as charging occurs during selected daytime hours and discharging is triggered mechanically, which can lead to fragmented charging and missed alignment with the most valuable price spikes. In contrast, nominal optimization coordinates decisions across time, producing smoother charging and sharper and more concentrated discharging during peak-price periods, more effectively capturing intraday price fluctuations.

Daily Case Study: December 24 Prescriptions

Hours	Action	Hourly Power (MW)
00–13	Hold	0.00
13–15	Charge	1.53
15–24	Hold	0.00

Table 3: **(a)** Rule-based heuristic

Hours	Action	Hourly Power (MW)
00–10	Hold	0.00
10–15	Charge	1.46
15–17	Hold	0.00
17–23	Discharge	1.13
23–24	Hold	0.00

Table 5: **(c)** Ellipsoidal robust

Hours	Action	Hourly Power (MW)
00–11	Hold	0.00
11–16	Charge	1.49
16–17	Hold	0.00
17–18	Discharge	4.04
18–24	Hold	0.00

Table 4: **(b)** Nominal optimization

Hours	Action	Hourly Power (MW)
00–09	Hold	0.00
09–15	Charge	1.04
15–17	Hold	0.00
17–23	Discharge	0.94
23–24	Hold	0.00

Table 6: **(d)** Budget robust

Table 7: December 24 dispatch prescriptions across strategies. Reported values are constant hourly power levels applied within each time block, the total energy moved over a block equals power multiplied by duration.

A one-day case study further clarifies these behavioral differences. Table 7 reports the December 24 prescriptions implied by each strategy. The rule-based heuristic charges briefly (hours 13–15) but does not discharge at all, leaving energy stranded overnight and failing to monetize the evening peak window. In contrast, the nominal optimizer follows a clear store-then-release pattern as it charges steadily through the midday period (hours 11–16) and executes a concentrated discharge during the peak hour (17–18), fully depleting the battery by the end of the day.

Both robust strategies preserve this qualitative structure of charging during the solar-rich daytime period and discharging during the evening but exhibit more conservative discharge intensity. The ellipsoidal robust solution remains close to nominal behavior, while the budgeted robust solution reduces cycling further by discharging at a lower hourly rate over an extended window (hours 17–23). This mirrors the weekly results where robustness trades some expected profit for protection against unfavorable price realizations, while still improving substantially over heuristic dispatch.

9 Discussion

Economic Value of Optimization

The results demonstrated that the primary economic gains from storage arise not just from the presence of a battery, but from coordinated, horizon-wide optimization of charging and discharging decisions. While rule-based heuristics capture broad daily price cycles and offer modest improvements over direct solar export, they remain inherently myopic. Additionally, fixed thresholds and local decision rules fail to account for intertemporal trade-offs in state of charge, often leading to stranded energy or poorly timed discharges.

In contrast, optimization-based dispatch internalizes future price expectations and state-of-charge dynamics, allowing the battery to concentrate discharge during the most valuable periods. This coordination produces more decisive actions, which results in higher revenue while without increasing total cycling dramatically. From an operational perspective, these findings suggest that the economic value of storage is driven as much by decision quality as by physical capacity, emphasizing the importance of optimization

frameworks for real-world deployment.

Impact of Solar Uncertainty

We did look to introduce uncertainty in solar generation, however it had a comparatively limited impact on dispatch behavior and realized profit. This outcome is largely attributable to the operating context that San Diego exhibits relatively stable and predictable solar production, particularly over short horizons. As a result, deviations from point forecasts are typically small and rarely binding relative to battery power or energy constraints.

Moreover, solar uncertainty primarily affects charging opportunities rather than revenue-generating discharge decisions, which are driven by price dynamics. Since the battery is frequently energy-limited rather than power-limited during peak periods, modest reductions in available solar generation do not materially alter optimal discharge behavior. These findings suggest that in regions with high solar reliability, modeling effort may be better allocated toward price uncertainty rather than generation uncertainty when designing robust dispatch strategies.

Price Robustness Tradeoffs

Incorporating price robustness introduces a clear trade-off between expected profit and protection against unfavorable prices. Both ellipsoidal and budgeted robust formulations preserve the qualitative structure of the nominal optimization solution of charging during solar-rich daytime hours and discharging during the evening peak but differ in the aggressiveness of their responses.

The ellipsoidal robust model remains close to the nominal solution, discharging more evenly without reducing total energy output. This behavior reflects a moderate hedging strategy that penalizes extreme price deviations without fully abandoning high-value opportunities. In contrast, the budgeted robust formulation enforces stricter protection by limiting the number and magnitude of unfavorable prices considered, leading to lower discharge intensity spread over longer periods.

Importantly, despite their conservatism, both robust strategies outperform heuristic dispatch, indicating that robustness does not negate the benefits of optimization. Instead, this robustness provides a tunable approach for balancing risk and return, allowing operators to select a dispatch policy aligned with their tolerance for price uncertainty and revenue volatility.

10 Limitations and Future Work

Limitations

Since our project focuses on a single market context and operating environment (San Diego & SDG&E), the generalizability of the results are limited, especially as San Diego’s solar generation is relatively stable and price dynamics exhibit strong daily structure. Dispatch behavior and the relative value of robustness may differ in regions with higher renewable volatility, different market designs, or more pronounced congestion effects.

In addition, the battery configuration is fixed throughout the analysis, with a single energy capacity and power rating. While this isolates the impact of dispatch strategy, it does not capture how optimal behavior may change across different battery sizes or power-to-energy ratios.

Future Extensions

There are several extensions that could build on this framework to better reflect real-world operating conditions. First, incorporating real-time or pricing changes within the hour, rather than relying solely on day-ahead hourly prices, would allow the model to capture additional revenue opportunities and short-term volatility. This would be particularly relevant for fast-responding storage assets.

Additionally, extending the optimization horizon beyond a fixed one-week window could enable more

strategic between-day energy shifting, especially during prolonged price trends or weather events. Longer horizons may also improve end-of-horizon state-of-charge decisions.

Lastly, instead of using a fixed cost for battery wear, the model could learn how the battery actually degrades over time and adjust decisions to trade off short-term profit against keeping the battery healthy longer. Second, adding simple risk penalties could help limit big swings in revenue, rather than only protecting against the absolute worst price scenarios. Together, these improvements would make the battery model more realistic and more useful in real-world operation.

11 Conclusion

Overall, this work demonstrates the value of integrating forecasting, optimization, and robustness in the operation of solar-coupled battery storage systems. By combining data-driven solar forecasting with horizon-wide dispatch optimization, the proposed framework substantially outperforms heuristic and no-storage baselines. This highlights that the economic value of storage is driven as much by coordinated decision-making as by physical capacity. Optimization-based dispatch enables the battery to anticipate price dynamics, allocate state of charge strategically, and concentrate discharging during the most profitable periods.

Robust planning further enhances this framework by explicitly accounting for price uncertainty. While robustness introduces a trade-off between expected profit and conservatism, both ellipsoidal and budgeted robust formulations preserve the core structure of the optimal dispatch and consistently outperform heuristic strategies. These results suggest that robustness serves as a practical risk-management tool rather than a substitute for optimization, allowing operators to tailor dispatch behavior to their tolerance for revenue volatility.

Overall, the findings underscore the importance of having end-to-end decision pipelines that link predictive modeling with prescriptive optimization. As electricity markets continue to experience increasing renewable penetration and price uncertainty, integrated approaches like this will be essential for unlocking the full economic potential of energy storage assets.

References

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